Two Ways to Build an Interval

AP Statistics · Unit 3 · Topic 3.10 · B: 2026-12-11 · E: 2027-01-22 · TEACHER KEY

CED TETHER

- **Skill 1.D** — Identify an appropriate inference method for confidence intervals.
- **Skill 4.C** — Verify that inference procedures apply in a given situation.
- **Skill 3.D** — Construct a confidence interval, provided conditions for inference are met.
- **EU UNC-4** — An interval of values should be used to estimate parameters, in order to account for uncertainty.
- **LO UNC-4.I** — Identify an appropriate confidence interval procedure for a comparison of population proportions. [Skill 1.D]
- **EK UNC-4.I.1** — The appropriate confidence interval procedure for a two-sample comparison of proportions for one categorical variable is a two-sample z-interval for a difference between population proportions.
- **LO UNC-4.J** — Verify the conditions for calculating confidence intervals for a difference between population proportions. [Skill 4.C]
- **EK UNC-4.J.1** — In order to calculate confidence intervals to estimate a difference between proportions, we must check for independence and that the sampling distribution is approximately normal: a. To check for independence: i. Data should be collected using two independent, random samples or a randomized experiment.
- ii. When sampling without replacement, check that n1 $\leq$ 10%N1 and n2 $\leq$ 10%N2.
- b. To check that the sampling distribution of $\widehat{p}_1$ - $\widehat{p}_2$ is approximately normal (shape): i. For categorical variables, check that n1*$\widehat{p}_1$, n1*(1-$\widehat{p}_1$), n2*$\widehat{p}_2$, and n2*(1-$\widehat{p}_2$) are all greater than or equal to some predetermined value, typically either 5 or 10.
- **LO UNC-4.K** — Calculate an appropriate confidence interval for a comparison of population proportions. [Skill 3.D]
- **EK UNC-4.K.1** — For a comparison of proportions, the interval estimate is ($\widehat{p}_1$ - $\widehat{p}_2$) $\pm$ z* · sqrt($\widehat{p}_1$(1-$\widehat{p}_1$)/n1 + $\widehat{p}_2$(1-$\widehat{p}_2$)/n2). (Note: these formulas can be constructed from the general formula and SE formulas on the AP formula sheet; they need not be memorized.)
- **LO UNC-4.L** — Calculate an interval estimate based on a confidence interval for a difference of proportions. [Skill 3.D]
- **EK UNC-4.L.1** — Confidence intervals for a difference in proportions can be used to calculate interval estimates with specified units.

Essential question: How do the study design and variance structure determine a valid interval for a difference in population proportions?

DOK-3 skill: 4.C · **FRQ pattern:** `two-sample-interval-vs-separate-intervals`

DOK rationale: Part (c) requires adjudicating two plausible interval methods by coordinating the two-sample design, condition evidence, variance structure, confidence-level target, and the inferential consequence of an invalid construction.

Do Now (first take) → Explore (video + follow-along) → Exit (finish + turn in) DOK: 1 → 3 **Minutes:** 45

Teacher does

- Board slide up at the bell. Ask students to choose a method before confirming it.
- Begin the video at minute 5; return at minute 33 to separate variance, SE, and margin of error.
- Collect at the door. Score part (c) only, E/P/I.

Students do

- Choose Elena or Devon and cite design, counts, or zero.
- Complete the follow-along, finish (a)–(c), revise, and turn in the sheet.

Questions to ask

- What single parameter is the town comparison trying to estimate?
- Which four observed counts justify the normal approximation?
- Do standard errors or variances add?

Adult role
Solo teacher. Circulate 5 pairs. Link the formula to design and the rejected method to zero.

[WARN] WATCH FOR

- Pooling the two sample proportions even though the task is to estimate an unrestricted difference.
- Adding separate margins of error instead of adding estimated variance terms under the square root.
- Assuming two individual 95 percent intervals automatically yield a 95 percent interval for their difference.

SCORING PART (c) — E / P / I

Expected elements

- **judgment-1** (required) — Chooses Elena using independent samples and adding variance contributions under one square root.
- **judgment-2** (required) — Explains that separate intervals are not the requested 95 percent procedure for the difference.
- **judgment-3** (required) — Uses Elena's positive interval versus Devon's interval containing zero to explain the changed inference.

E	Meets all three checks with correct contextual reasoning; equivalent wording is acceptable.
P	Shows at least one substantive correct check, but has an omission or error that prevents E.
I	Shows none of the three checks with substantive correct reasoning; a choice alone is insufficient.

[STAR] TODAY'S PROBLEM — annotated

Town Oak has 2,200 households and Pine 1,800. Independent SRSs find 77 of 140 Oak households and 48 of 120 Pine households use curbside composting. Let $p_O - p_P$ mean Oak minus Pine.

Elena: "The two-sample z-interval is $(0.030, 0.270)$."

Devon: "Subtract the farthest endpoints of separate 95% intervals. That gives $(-0.020, 0.320)$, so zero is plausible."

Household samples

Town	Use	Other	n	N
Oak	77	63	140	2200
Pine	48	72	120	1800

FIRST TAKE (5 min)

The two reported ranges disagree about whether the towns could be equal. Which range would you trust, and why?
Key: Not graded. Equal centers do not make both uncertainty calculations valid.

Rules callout "BUILD ONE INTERVAL FOR THE DIFFERENCE (keep on the page)" is printed on the student sheet.

DOK 1 **(a)** Define p_O and p_P and name the interval procedure for $p_O - p_P$.

Key: p_O and p_P are the proportions of all Oak and Pine households using their town's service. Use a two-sample z-interval for $p_O - p_P$.

DOK 2 **(b)** Check independent samples, both 10 percent conditions, and all four counts. Compute the estimate, SE, margin of error, and 95% interval.

Key: The SRSs are independent; $140 \leq 220$ and $120 \leq 180$; and counts 77, 63, 48, and 72 exceed 10. Thus $\hat{p}_O - \hat{p}_P = 0.55 - 0.40 = 0.15$, $SE = 0.0613829$, and $ME = 1.96(0.0613829) = 0.1203104$. The interval is $(0.0296896, 0.2703104) \approx (0.030, 0.270)$.

DOK 3 **(c)** In 3–4 sentences, choose a method using the independent samples and the variance formula. Explain why subtracting separate 95% intervals does not establish a 95% interval for the difference. Use zero's location to explain how the method changes the conclusion.

Key: Elena is warranted. Independent SRSs and counts 77, 63, 48, and 72 support one two-sample interval. Each sample proportion estimates its own population proportion; independence makes the two estimated variance terms add under one square root. Devon instead combines separate intervals $(0.4676, 0.6324)$ and $(0.3123, 0.4877)$ into $(-0.0201, 0.3201)$. Two individual 95% intervals do not guarantee one 95% interval for their difference. His wider result includes zero and falsely makes no difference look plausible; Elena's requested interval supports a positive Oak-minus-Pine difference of about 3 to 27 points.

EXIT

One thing the video changed about my first take: ______________________________